2025 WSMO Accuracy Round Solutions

SMO Team

AR 1: Real numbers x and y satisfy

$$x + y = 10 \quad \text{and} \quad x^2 + y^2 = 58.$$

Find $|x - y|$.

Answer: $\boxed{4}$

Solution: We have

$$(x-y)^2 = 2(x^2+y^2) - (x+y)^2 = 116 - 100 = 16 \implies |x-y| = \boxed{4}.$$

AR 2: Points A, B, C, D, and E are collinear such that $AB = 20$, $BC = 52$, $CD = 34$, and $DE = 25$. Find the minimum possible length of EA.

Answer: $\boxed{13}$

Solution: Place the points on a number line with $A = 0$. Then, $E = \pm 20 \pm 52 \pm 34 \pm 25$ for some choice of signs, so EA is the positive difference between the sums of two groups that partition $\{20, 52, 34, 25\}$. Since $20 + 52 + 34 + 25 = 131$, we want a group with a sum as close to $\frac{131}{2}$ as possible. The closest possible sums are $34 + 25 = 59$ and $20 + 52 = 72$, giving a minimum of $72 - 59 = \boxed{13}$, which is achieved by $A = 0$, $B = 20$, $C = 72$, $D = 38$, and $E = 13$.

AR 3: On a string of holiday lights, there are 4 blue lights for every 3 green lights, 7 red lights for every 6 blue lights, and 20 white lights for every 13 colored lights. What is the smallest number of lights that can be on the string of lights?

Answer: $\boxed{1155}$

Solution: Since there are 4 blue lights for every 3 green lights and 7 red lights for every 6 blue lights, the numbers of green, blue, and red lights are in a $9 : 12 : 14$ ratio. Let there be $9k$ green, $12k$ blue, and $14k$ red lights, for a total of $35k$ colored lights. Then, there are $\frac{20}{13} \cdot 35k$ white lights, so $13 \mid k$. The number of lights is minimized when $k = 13$, giving

$$35 \cdot 13 + 20 \cdot 35 = 455 + 700 = \boxed{1155}.$$

AR 4: Evaluate:

$$\sum_{1 \le a,b \le 25} \left\lfloor \frac{a + b\sqrt{2}}{1 + \sqrt{2}} \right\rfloor.$$

Answer: $\boxed{7825}$

Solution: Note that

$$\frac{a+b\sqrt{2}}{1+\sqrt{2}}+\frac{b+a\sqrt{2}}{1+\sqrt{2}}=\frac{(a+b)(1+\sqrt{2})}{1+\sqrt{2}}=a+b.$$

In addition,

$$\frac{a+b\sqrt{2}}{1+\sqrt{2}}=(a+b\sqrt{2})(\sqrt{2}-1)=(2b-a)+(a-b)\sqrt{2},$$

which is irrational when $a \neq b$. So, if $a \neq b$, the floors of the terms for (a, b) and (b, a) sum to $a+b-1$, and if $a=b$, the term is exactly a. There are 600 ordered pairs with $a \neq b$, and

$$\sum_{a\neq b}(a+b)=\sum_{1\leq a,b\leq 25}(a+b)-2\sum_{a=1}^{25}a=2\cdot 25\cdot 325-2\cdot 325=15600.$$

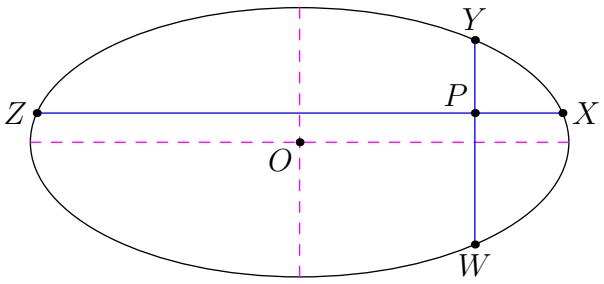
Thus, the sum is

$$\frac{1}{2}\sum_{a\neq b}(a+b-1)+\sum_{a=1}^{25}a=\frac{15600-600}{2}+325=\boxed{7825}.$$

AR 5: Let $\mathcal{E}$ be an ellipse. Points W, X, Y, and Z lie on $\mathcal{E}$ such that WY and XZ are parallel to the minor and major axes of $\mathcal{E}$, respectively. Suppose that segments WY and XZ intersect at point P. Given that $PW=9$, $PX=6$, $PY=5$, and $PZ=30$, the area of $\mathcal{E}$ can be expressed as $m\pi$. Find m.

Answer: $\boxed{170}$

Solution:



Let $\mathcal{E}$ be centered at the origin O with equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, where $a>b$, so XZ is horizontal and WY is vertical. Since XZ is a horizontal chord, its midpoint lies on the y-axis, so the distance from P to the y-axis is $\frac{30-6}{2}=12$. Similarly, the distance from P to the x-axis is $\frac{9-5}{2}=2$. WLOG, let $P=(12,2)$, so $X=(18,2)$ and $Y=(12,7)$. Since X and Y lie on $\mathcal{E}$, we have

$$\frac{324}{a^2}+\frac{4}{b^2}=1 \quad \text{and} \quad \frac{144}{a^2}+\frac{49}{b^2}=1.$$

Solving, we have $a^2=340$ and $b^2=85$, so the area of $\mathcal{E}$ is

$$\pi ab=\pi\sqrt{340\cdot 85}=170\pi \implies \boxed{170}.$$

AR 6: Let $t_1, t_2, \ldots, t_6$ be integers and $N < 100$ be a positive integer such that $Nt_1 < 60t_2 < 27t_3 < 8t_4 < 50t_5 < 45t_6$. Over all choices of $t_1, t_2, \ldots, t_6$, the minimum possible value of $45t_6 - Nt_1$ is 25. Find the sum of all possible values of N.

Answer: $\boxed{80}$

Solution: Let $v_1 = Nt_1$, $v_2 = 60t_2$, $v_3 = 27t_3$, $v_4 = 8t_4$, $v_5 = 50t_5$, and $v_6 = 45t_6$. Since v_2 and v_6 are both multiples of 15, $v_6 - v_2$ is a positive multiple of 15. We first find when $v_6 - v_2 = 15$. Let $v_2 = 60a$. Since $45 \mid 60a + 15$, we have $a \equiv 2 \pmod 3$. Since $v_6 - v_5$ is a positive multiple of 5 and $v_5 - v_2$ is a positive multiple of 10, we must have $v_5 = v_2 + 10$, so $50 \mid 60a + 10 \implies a \equiv 4 \pmod 5$. Now, $v_3 - v_2$ is a positive multiple of 3 and $v_4 - v_3 \geq 1$, while $v_5 - v_4$ is a positive multiple of 2, so $v_3 \in \{v_2+3, v_2+6\}$. If $v_3 = v_2+3$, then $27 \mid 60a + 3$ gives $a \equiv 4 \pmod 9$, which contradicts $a \equiv 2 \pmod 3$. So, $v_3 = v_2 + 6$, meaning $27 \mid 60a + 6 \implies a \equiv 8 \pmod 9$, and since $v_5 - v_4$ is even, $v_4 = v_2 + 8$, meaning $8 \mid 60a + 8 \implies a \equiv 0 \pmod 2$. Combining, we have $a \equiv 44 \pmod{90}$, so $v_6 - v_2 = 15$ if and only if

$$v_2 \equiv 60 \cdot 44 = 2640 \pmod{5400},$$

which is achieved by $(v_2, v_3, v_4, v_5, v_6) = (2640, 2646, 2648, 2650, 2655)$. Now, if $v_6 - v_2 \geq 30$, then $v_6 - v_1 > 25$. So, we need $v_6 - v_2 = 15$ and the minimum possible value of $v_2 - v_1$ to be 10. Let $d = \gcd(N, 5400)$. Since $v_2 - v_1 = 2640 + 5400k - Nt_1$ for integers k and t_1, the possible values of $v_2 - v_1$ are exactly the positive integers congruent to 2640 modulo d. If $d \mid 2640$, the smallest is d, so $d = 10$. Otherwise, the smallest is the remainder when 2640 is divided by d, so $d \mid 2630 = 2 \cdot 5 \cdot 263$, meaning $d \mid 10$, which contradicts the remainder being 10. Thus, we need $\gcd(N, 5400) = 10$, so $N = 10k$ for some k relatively prime to 2, 3, and 5. Since $N < 100$, we have $N = 10, 70$, giving an answer of $10 + 70 = \boxed{80}$.

AR 7: Let S be the set of all non-congruent right triangles with integer side lengths such that the area of the triangle is equal to three times the perimeter of the triangle. Find the sum of the lengths of hypotenuses of all the right triangles in S.

Answer: $\boxed{267}$

Solution: Let the legs be $a \leq b$ and the hypotenuse be c. Since the area is 3 times the perimeter, the inradius is $\frac{2 \cdot \text{area}}{\text{perimeter}} = 6$. Since the inradius of a right triangle is $\frac{a+b-c}{2}$, we have $c = a + b - 12$. Substituting into $a^2 + b^2 = c^2$, we have

$$a^2 + b^2 = (a + b - 12)^2 \implies ab - 12a - 12b + 72 = 0 \implies (a - 12)(b - 12) = 72.$$

If $a - 12$ and $b - 12$ are both negative, then $(12 - a)(12 - b) = 72$ with $12 - a, 12 - b \leq 11$, which forces $(a, b) = (3, 4)$, but then $c = -5$. So,

$$(a - 12, b - 12) = (1, 72), (2, 36), (3, 24), (4, 18), (6, 12), (8, 9),$$

giving hypotenuses $c = a + b - 12 = 85, 50, 39, 34, 30, 29$. Thus, the answer is

$$85 + 50 + 39 + 34 + 30 + 29 = \boxed{267}.$$

AR 8: Nonnegative real numbers x and y satisfy the system of equations

$$16x^6 - 24x^4 + 9x^2 = \frac{y+1}{2}$$
$$8y^3 - 6y = 1.$$

The smallest possible value of $\arccos(x)$ can be expressed as $\frac{m}{n}^\circ$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{13}$

Solution: From the triple angle formula $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$, the roots of $8y^3 - 6y = 1$ are $\cos 20^\circ$, $\cos 100^\circ$, and $\cos 140^\circ$. Since $y \geq 0$, we have $y = \cos 20^\circ$, so

$$\frac{y+1}{2} = \frac{1+\cos 20^\circ}{2} = \cos^2 10^\circ.$$

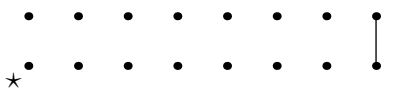
Now, note that $16x^6 - 24x^4 + 9x^2 = (4x^3 - 3x)^2$. If $x > 1$, then $4x^3 - 3x - 1 = (x-1)(2x+1)^2 > 0$, so the left hand side is greater than 1. So, $0 \leq x \leq 1$, and we may let $x = \cos\varphi$ for $0^\circ \leq \varphi \leq 90^\circ$. Then, the first equation becomes

$$\cos^2(3\varphi) = \cos^2(10^\circ) \implies \cos(3\varphi) = \pm\cos 10^\circ.$$

The smallest such φ is when $3\varphi = 10^\circ$, so the smallest possible value of $\arccos(x)$ is

$$\frac{10^\circ}{3} \implies 10 + 3 = \boxed{13}.$$

AR 9: Amy the ant lives in the 8×2 rectangular grid of dots shown. From some dot D, she can move to a dot that is either horizontally adjacent to D, diagonally adjacent to D, or connected to D by a segment. In how many different ways can Amy visit every dot exactly once if she starts at the starred dot?



Answer: $\boxed{640}$

Solution: Number the columns $1, 2, \ldots, 8$ from left to right. Note that every move goes between adjacent columns, except for the move along the segment in column 8. Consider the order of the columns that Amy visits. Whenever Amy enters a column for the first time, she can move to either of its two dots, and when she enters it for the second time, her move is forced. Since the dot in column 1 is fixed, each order of columns corresponds to $2^7 = 128$ paths. Now, suppose Amy has visited both dots in each of the first $c-1$ columns and has just arrived at column c for the first time. She must move to column $c+1$. If she then returns to column c, she must move back to column $c+1$, after which she has visited both dots in each of the first $c+1$ columns. Otherwise, the remaining dot in column c must be the last dot she visits, so she must travel right to column 8 (turning back earlier would leave the columns to her right unreachable), use the segment, and travel back left to column c. Starting with $c = 1$, Amy can travel back to column c for $c = 1, 3, 5, 7$, or never travel back and end in column 8, giving 5 orders of columns. So, the answer is $5 \cdot 128 = \boxed{640}$.

AR 10: Let $f : \{1, 2, \ldots, 20\} \to \{1, 2, \ldots, 20\}$ be a function such that for all $m, n \in \{1, 2, \ldots, 20\}$, we have

$$\gcd(m, f(n)) = \gcd(n, f(m)).$$

How many such functions f are there?

Answer: $\boxed{960}$

Solution: Setting $m = f(n)$, we have $f(n) = \gcd(f(n), f(n)) = \gcd(n, f(f(n)))$, so $f(n) \mid n$ for all n. Now, fix a prime p, and let $g(n) = v_p(f(n))$, so $g(n) \leq v_p(n)$. Taking v_p of both sides of the given equation, we have

$$\min(v_p(m), g(n)) = \min(v_p(n), g(m))$$

for all m, n. Let t be the maximum value of $g(n)$, achieved at $n = n_0$. Then, for all m, we have

$$\min(v_p(m), t) = \min(v_p(n_0), g(m)) = g(m),$$

since $g(m) \leq t \leq v_p(n_0)$. So, $v_p(f(m)) = \min(v_p(m), t)$ for all m. Conversely, for any choice of t, this satisfies the equation, since both sides become $\min(v_p(m), v_p(n), t)$. Thus, $f(n) = \gcd(n, M)$ for some M that divides $\operatorname{lcm}(1, 2, \ldots, 20) = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$, and each such M gives a different function. So, the answer is the number of divisors of $\operatorname{lcm}(1, 2, \ldots, 20)$, which is

$$5 \cdot 3 \cdot 2^6 = \boxed{960}.$$